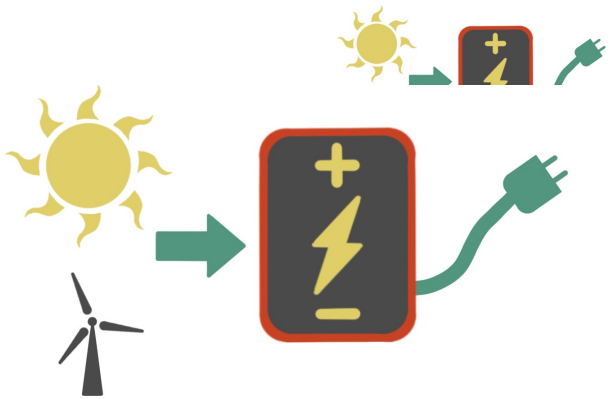
BMS State Estimation Algorithms



Ania's BMS Course

by Ania Mitros, PhD

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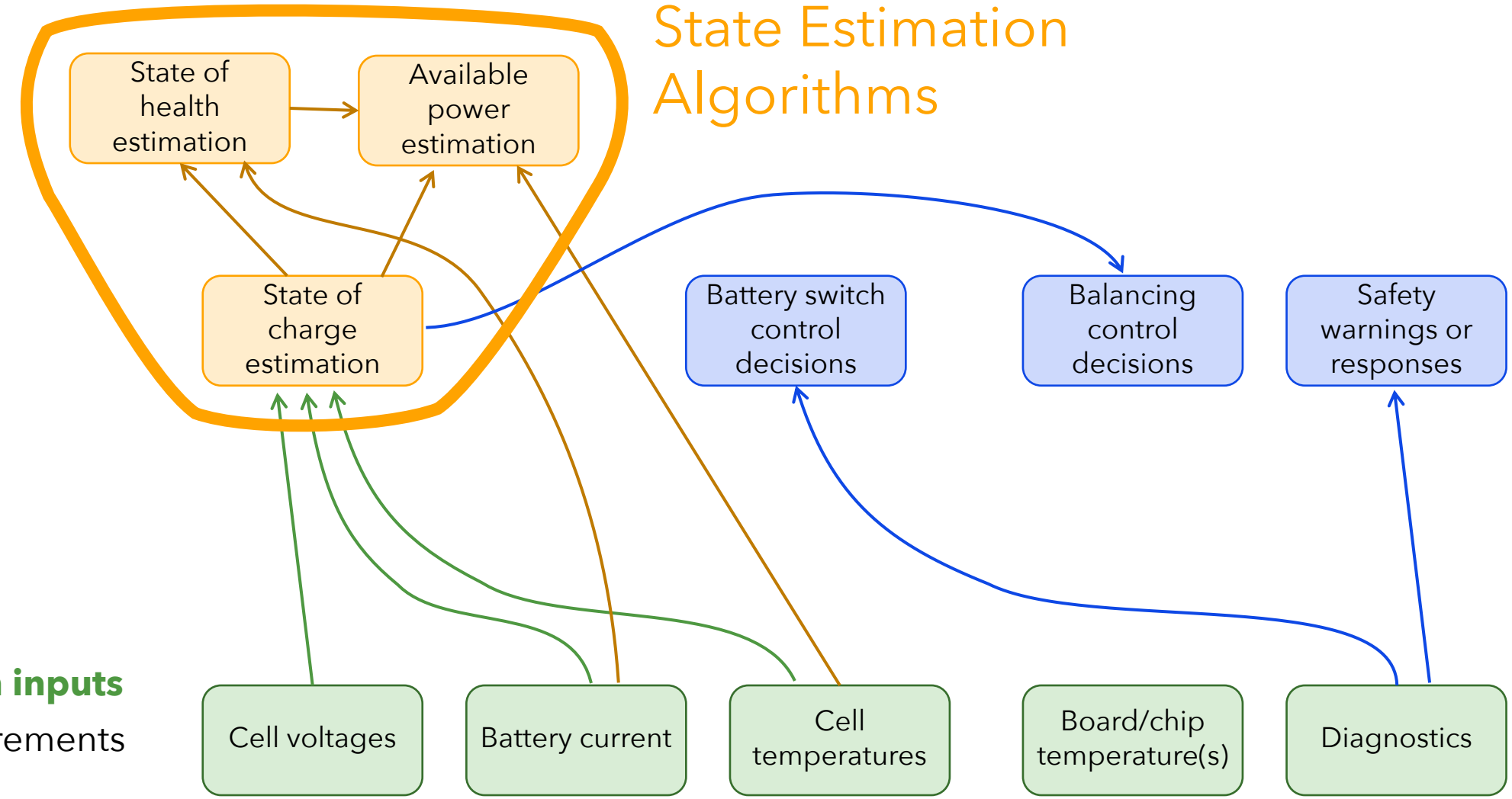
https://ofb.net/~ania/teaching/BMS-course/

BMS algorithms overview



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BMS State Estimation Algorithms

Just the basics.

- Overview
- State of charge
- State of health
- Power estimation

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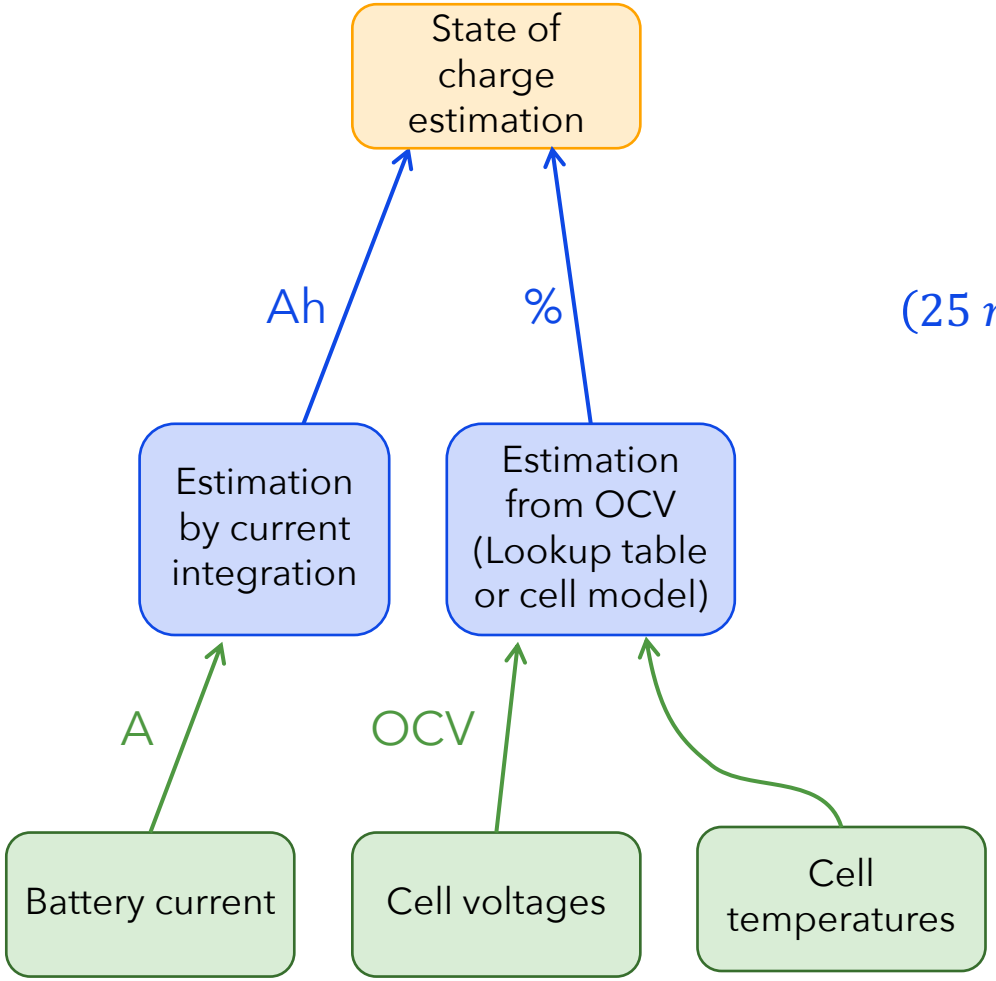
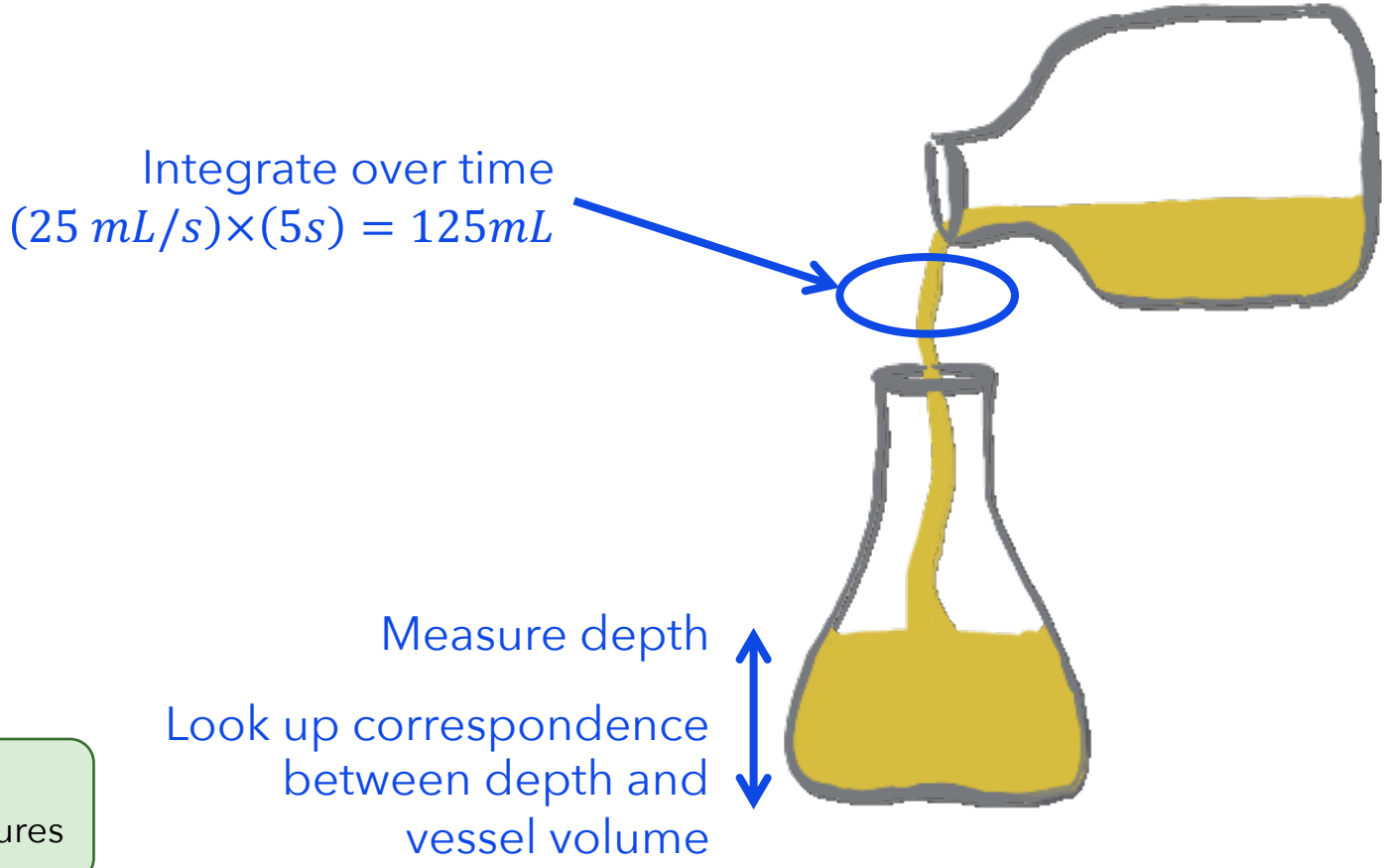
State of charge estimation





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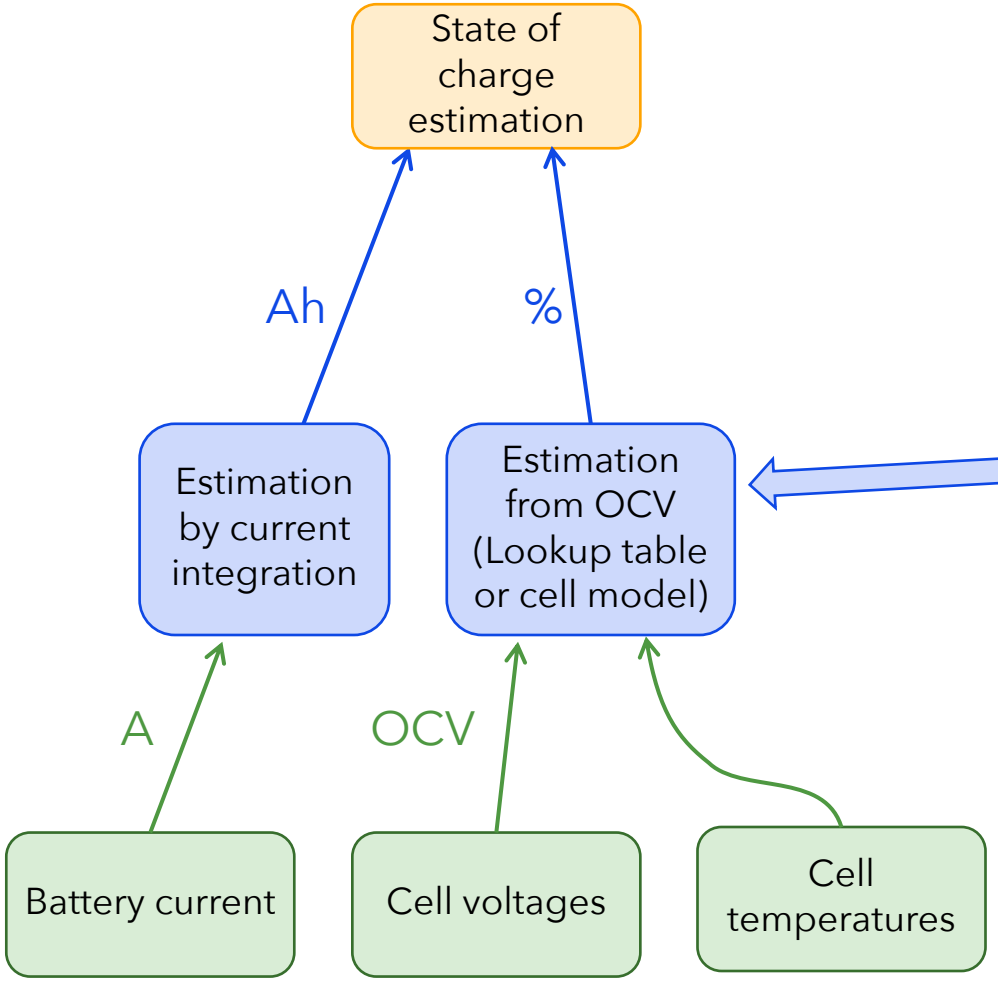
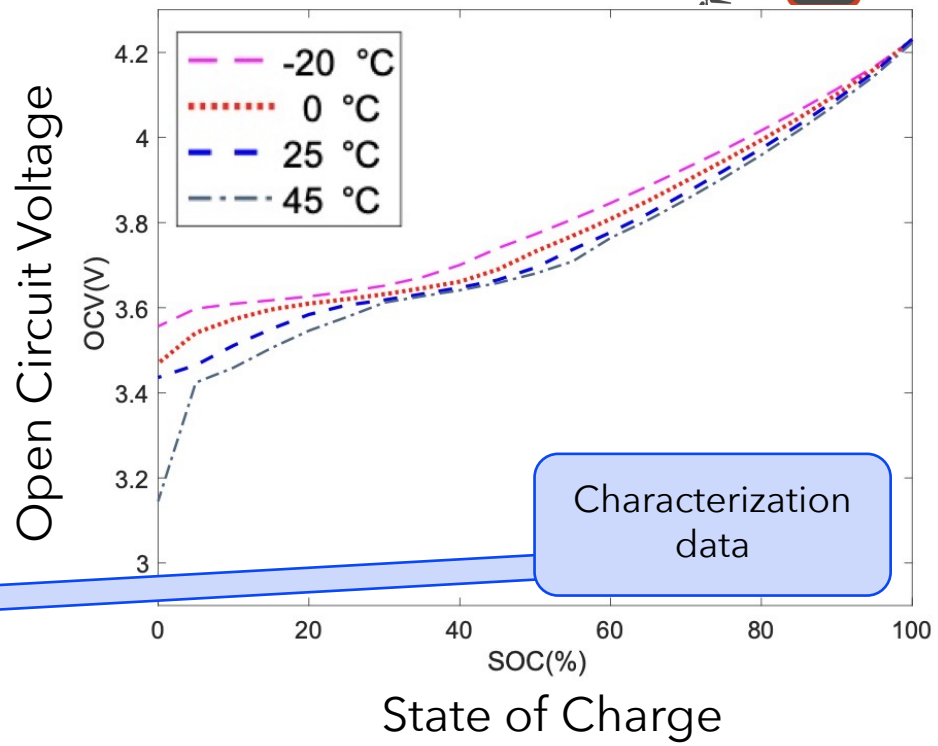
State of charge estimation

SOC is expressed in percent (%) or units of charge (Ah)





Plot from: Zhang 2018, doi:10.3390/en11092408

Algorithms:

- Estimation by current integration
- Estimation from OCV
- Combining both simply
- Combining both with Kalman filters

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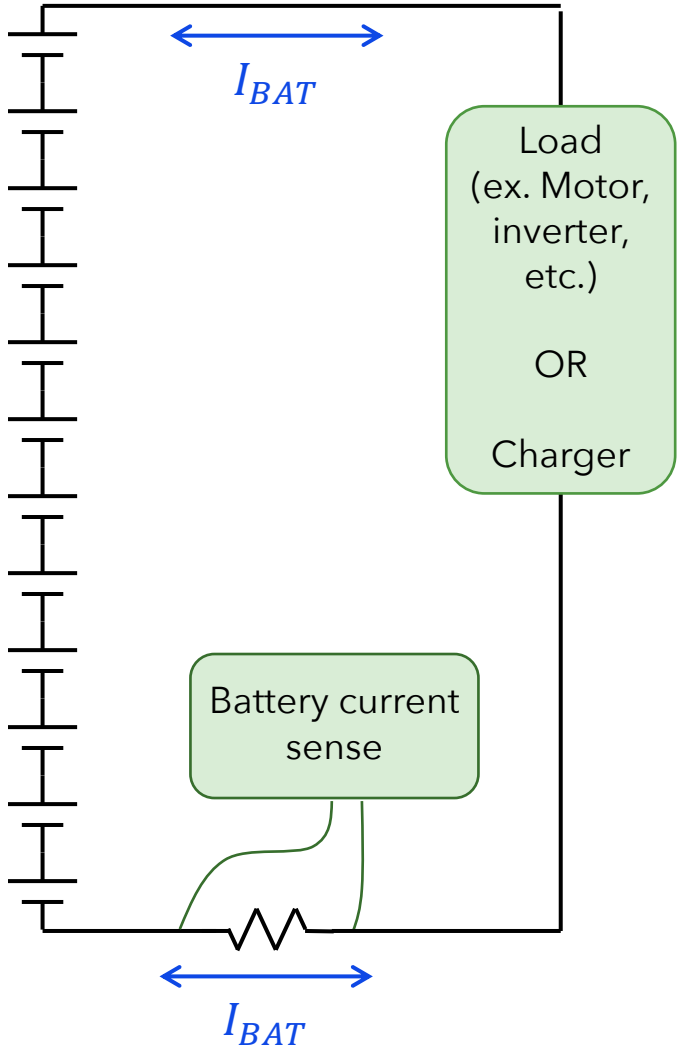
State of charge: Current integration



$$Q_{BAT} = \int I_{BAT} \cdot dt$$

Pro: Good method when current is flowing (when SOC estimation from voltage is tricky).

Con: Error accumulates without limit. Must have another method to reset to ground truth.

Ex.: INA240 ultra-precise current sense amplifier

Offset:

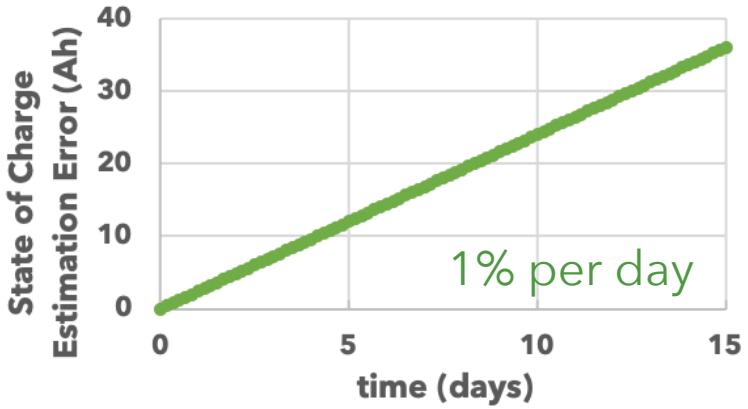
- **Offset Voltage: ±25 µV (Maximum)** $\longrightarrow$ $I_{OFFSET} = \frac{V_{OFFSET}}{250\mu\Omega} = 100mA$

$Q_{TOTAL} = 200\ Ah$

$I_{CONT} = 100\ A$

$R_{SHUNT} = 250\ \mu\Omega$

$P_{SHUNT} = I_{BAT}^2 \times R_{SHUNT} = 2.5\ W$



https://www.ti.com/product/INA240

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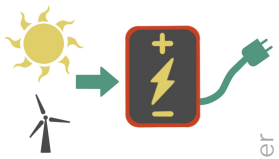
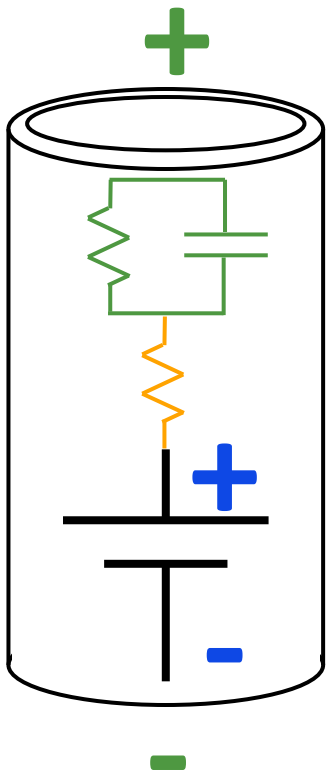
Review: cell voltage behaviors

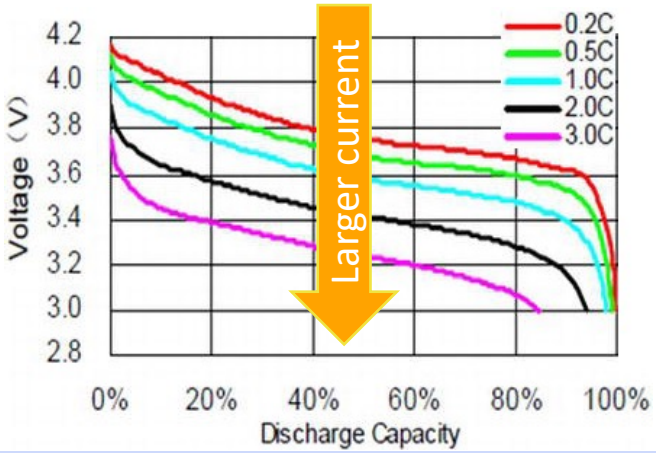


Terminal Voltage



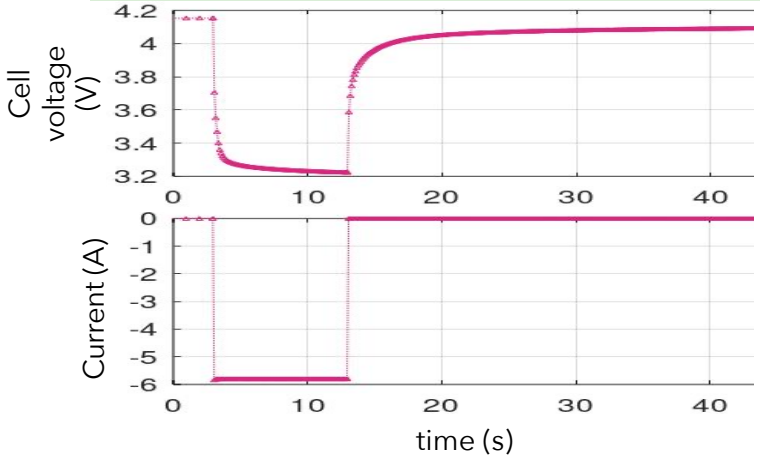
Open Circuit Voltage (OCV)

Cell impedance



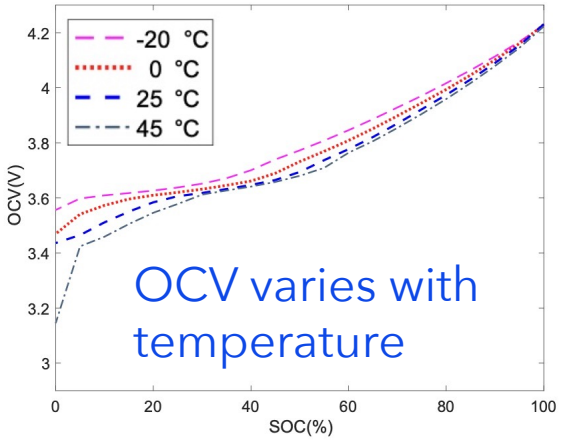
Temperature dependence: http://dx.doi.org/10.5772/intechopen.99851

Time-dependent response



Time-dependent data: Dr. Phillip Kollmeyer https://data.mendeley.com/datasets/wykht8y7tg/1#file-57a2002f-d618-4dd8-96e3-37ffb6bc26c5

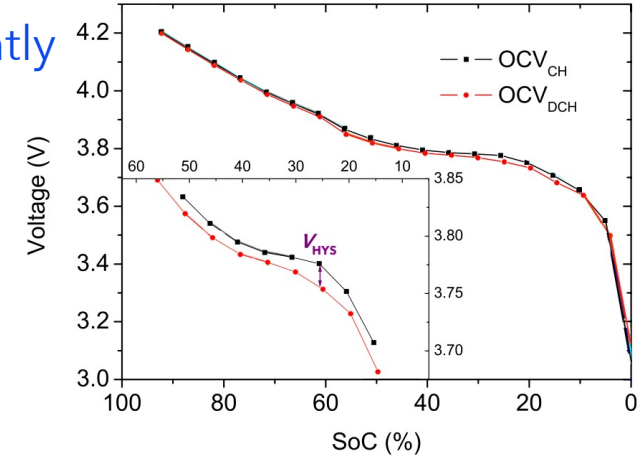
OCV varies with SOC and temperature (time-independent)



OCV varies with temperature

OCV vs temp: Zhang 2018, doi:10.3390/en11092408

OCV slightly differs in charge vs discharge



Hysteresis plot: Ovejas 2019, https://doi.org/10.1038/s41598-019-51474-5

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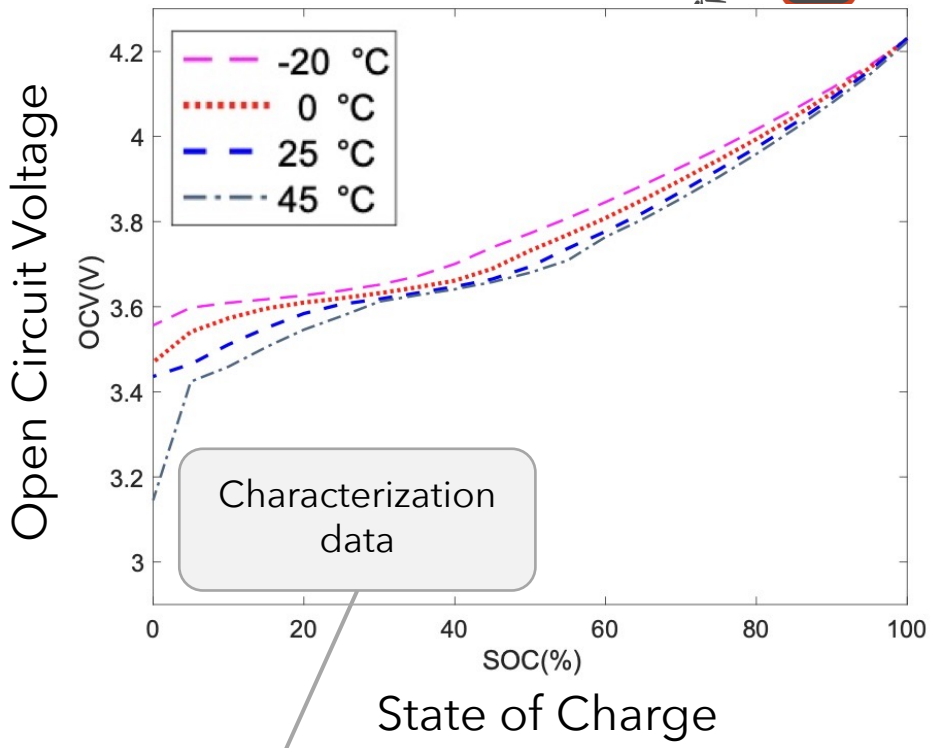
State of charge: OCV

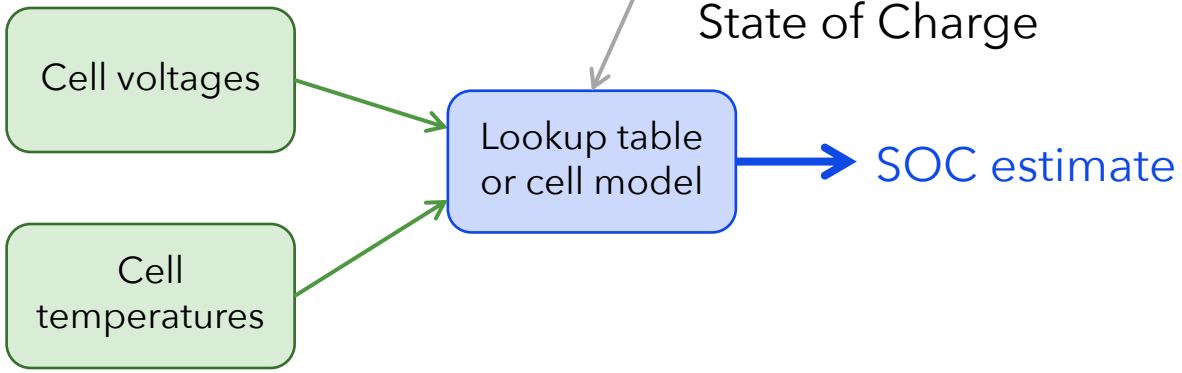
Pro: Error does not accumulate.

Cons:
- Tricky when current is flowing; OCV can only be measured with no current and a rested battery.
- Gives SOC %, so for aged reduced-capacity cells we need more information to get Ah.

→ Sufficient for some applications



Plot from: Zhang 2018, doi:10.3390/en11092408



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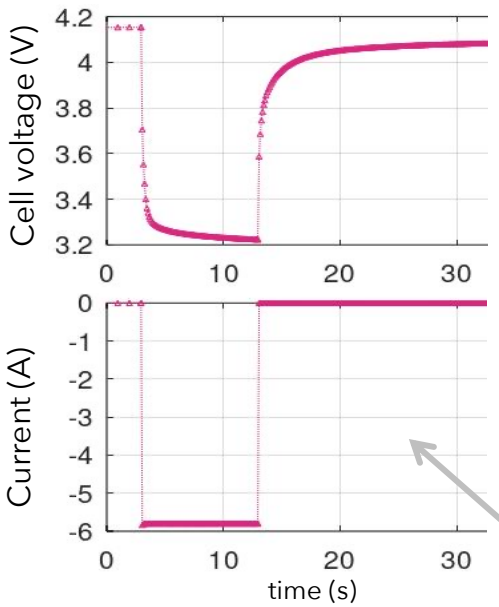
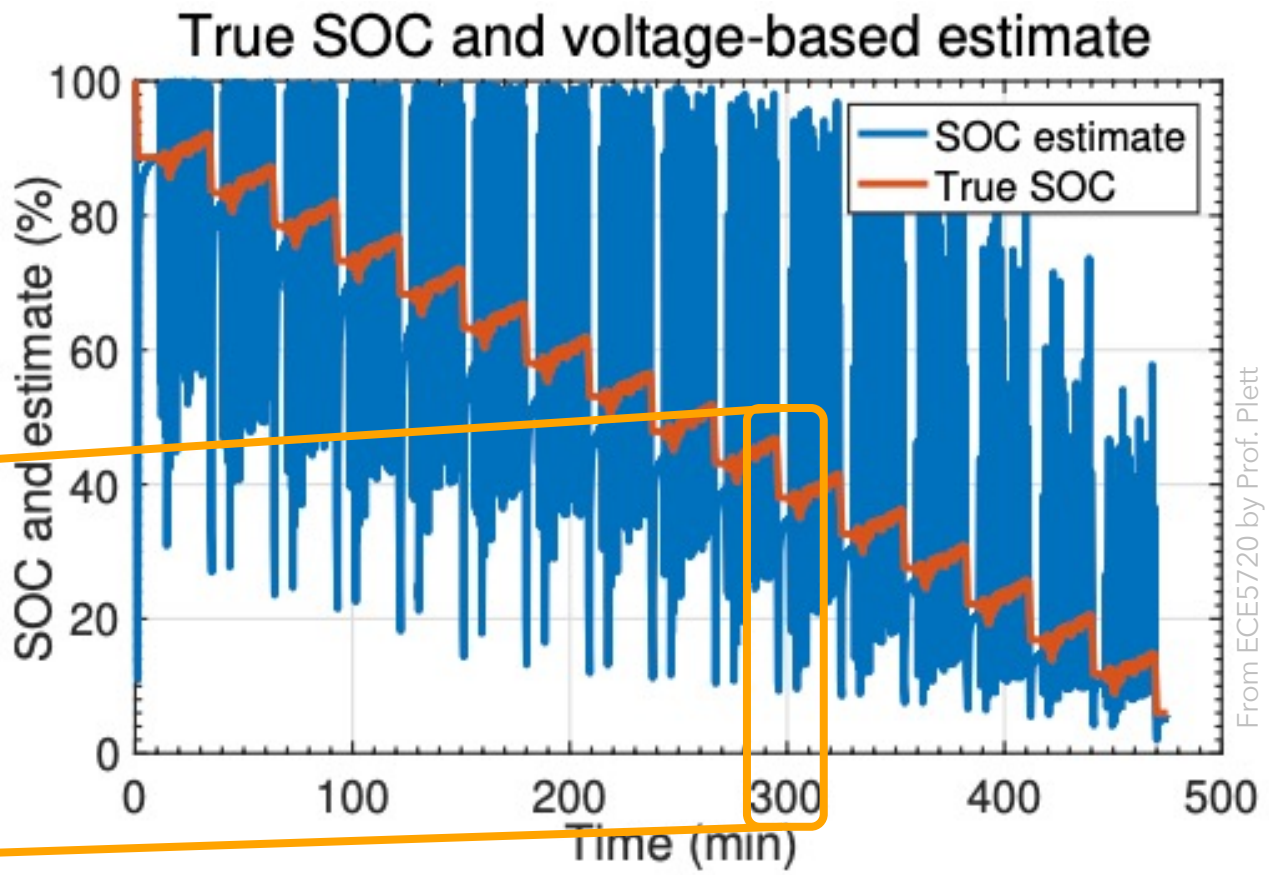
State of charge: Example of SOC from V_{CELL}

Algorithm:

- $v(t)$ is the measured cell voltage
- Assume:

  $$OCV(z(t)) = v(t)$$
- Look up $OCV(z(t))$ on SOC vs OCV curve
- Ignore cell impedance
- Ignore diffusion voltages and hysteresis





From ECE5720 by Prof. Plett
Univ. of Colorado, Colorado Springs
http://mocha-java.uccs.edu/ECE5720/

Data from Dr. Phillip Kollmeyer (phillip.kollmeyer@gmail.com)
https://data.mendeley.com/datasets/wykht8y7tg/1#file-57a2002f-d618-4dd8-96e3-37ffb6bc26c5

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Current integration vs OCV

For state of charge (SOC) estimation

Pause & Think

Current integration	OCV
Pro: Good method when current is flowing (when SOC estimation from voltage is tricky).	Pro: Error does not accumulate.
Con: Error accumulates without limit. Must have another method to reset to ground truth.	Cons: • Tricky with current flow; OCV can only be measured with no current and a rested battery. • Gives SOC %, so for aged reduced-capacity cells we need more information to get Ah.

1) One works during current, other works without current.

2) OCV can provide the reset!

3) Current integration provides Ah!

→ Complementary measurements!

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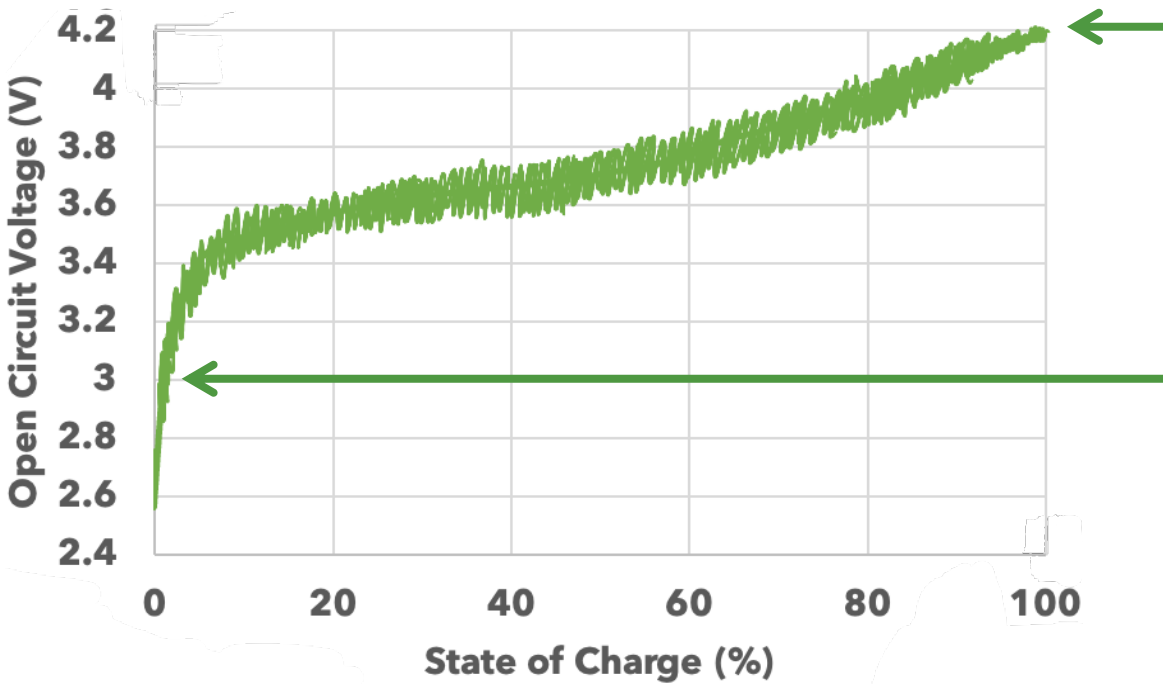
A simple sensor fusion algorithm




```
if any cell voltage ≥ Vmax:
        Stop charging
        SOC = 100%
```

```
if any cell voltage ≤ Vmin:
        Stop discharging
        SOC = 0%
```

```
while true:
        SOC estimate by current integration
```

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2nd simple sensor fusion algorithm

Charging	Discharging	Resting

SOC by current integration	SOC by OCV

N hours

24 hours

```
while true:
      SOC estimate by current integration
       if (current==0 for >N hours)
             OCV = cell voltage; # Approximation
             Reset SOC estimate by OCV
```

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Model-based sensor fusion: SOC from $V_{CELL} + I_{BAT} \times R$

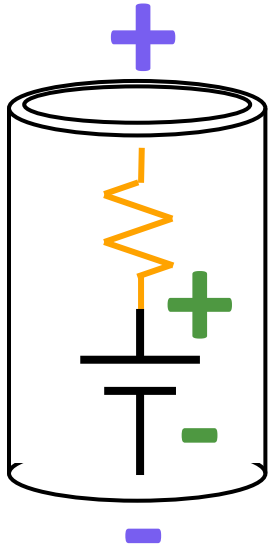
From ECE5720 by Prof. Plett
Univ. of Colorado, Colorado Springs
http://mocha-java.uccs.edu/ECE5720/

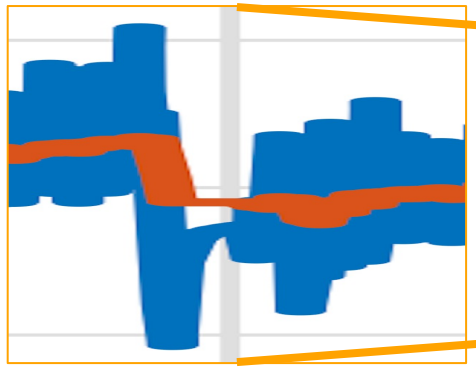
Algorithm:

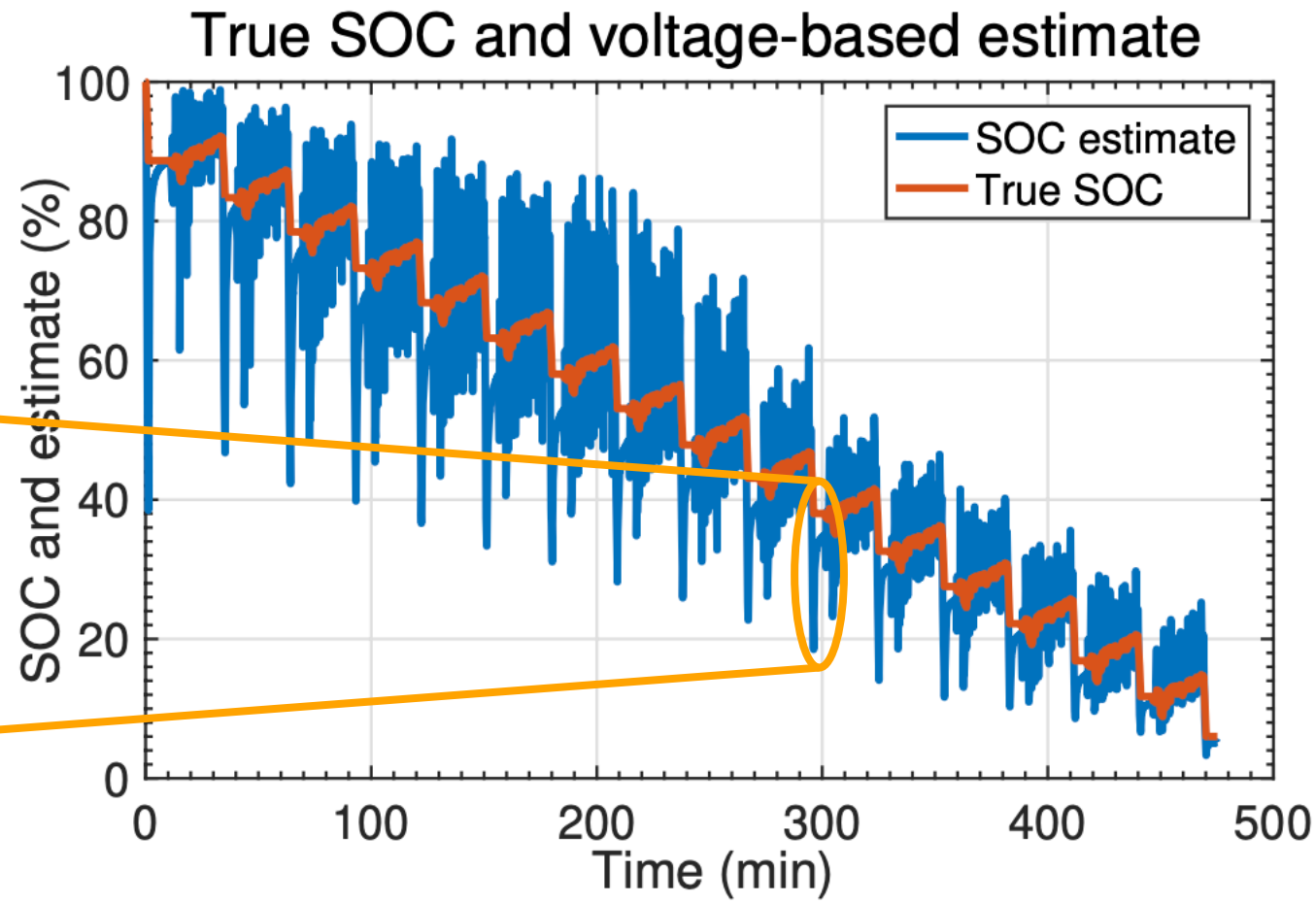
- $v(t)$ is the measured cell voltage
- Assume cell model:

$$\boxed{OCV(z(t)) = v(t) - i(t) \cdot R_0}$$

- Look up $OCV(z(t))$ on SOC vs OCV curve
- Ignore diffusion voltages and hysteresis
- Filtering would helps but add delay







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Better algorithms for SOC estimation

Model-based, iterative methods for estimating unobservable state variables (ex. state of charge) from observable variables (ex. voltage, current, temperature) in presence of process and system noise.

Measurements z_k → Matrix math → State estimate $\hat{x}_k$

at time k

Hat ^ indicates estimate

$$\begin{bmatrix} V_k \\ I_k \\ T_k \end{bmatrix} \times \begin{bmatrix} \blacksquare & \blacksquare & \blacksquare \end{bmatrix} \times \begin{bmatrix} \blacksquare & \blacksquare & \blacksquare \\ \blacksquare & \blacksquare & \blacksquare \\ \blacksquare & \blacksquare & \blacksquare \end{bmatrix} \times \ldots = \begin{bmatrix} \widehat{SOC}_k \end{bmatrix}$$

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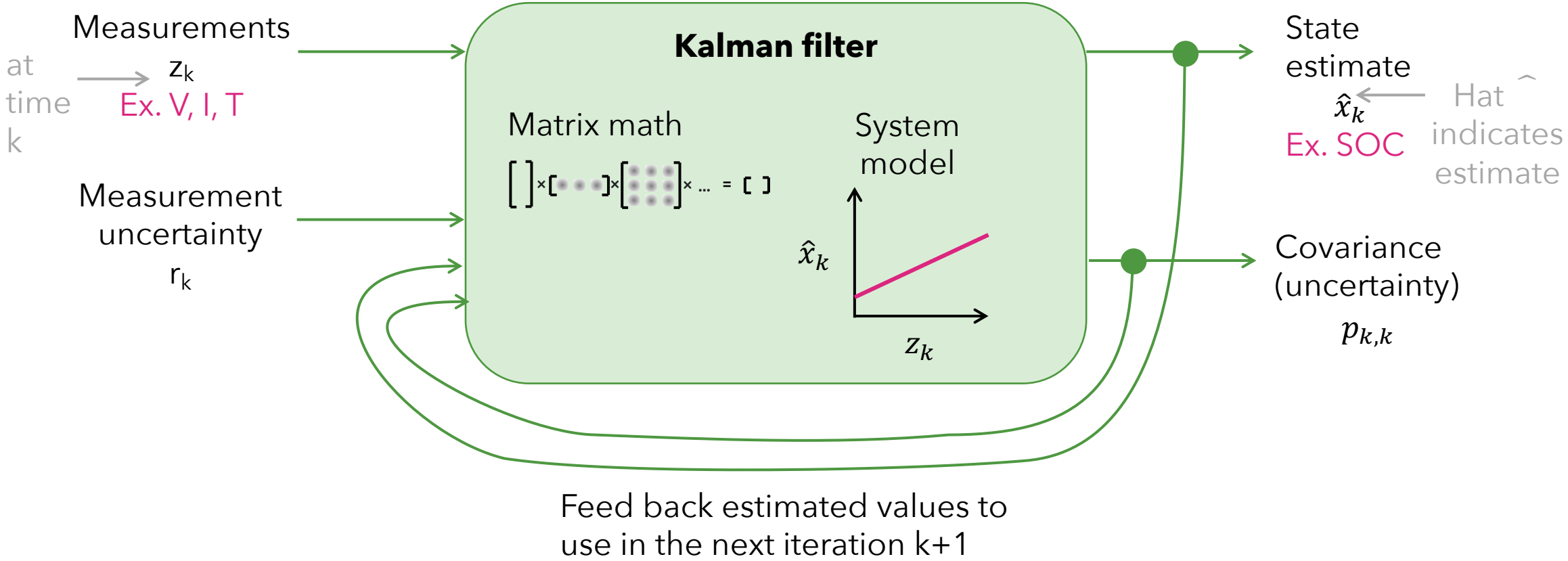
Better algorithms for SOC estimation

Model-based, iterative methods for estimating unobservable state variables (ex. state of charge) from observable variables (ex. voltage, current, temperature) in presence of process and system noise.



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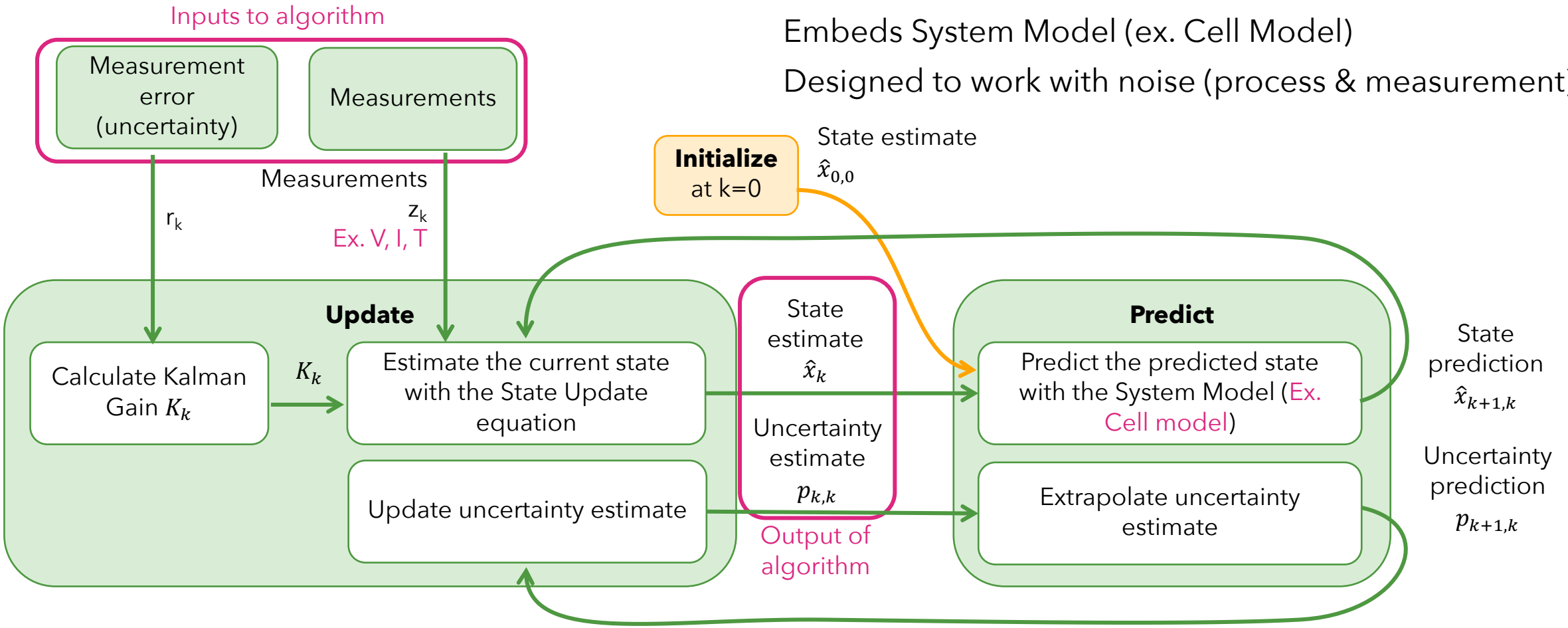
Kalman filter

The Kalman filter is:

Iterative

Embeds System Model (ex. Cell Model)

Designed to work with noise (process & measurement)



With gratitude to: https://www.kalmanfilter.net/

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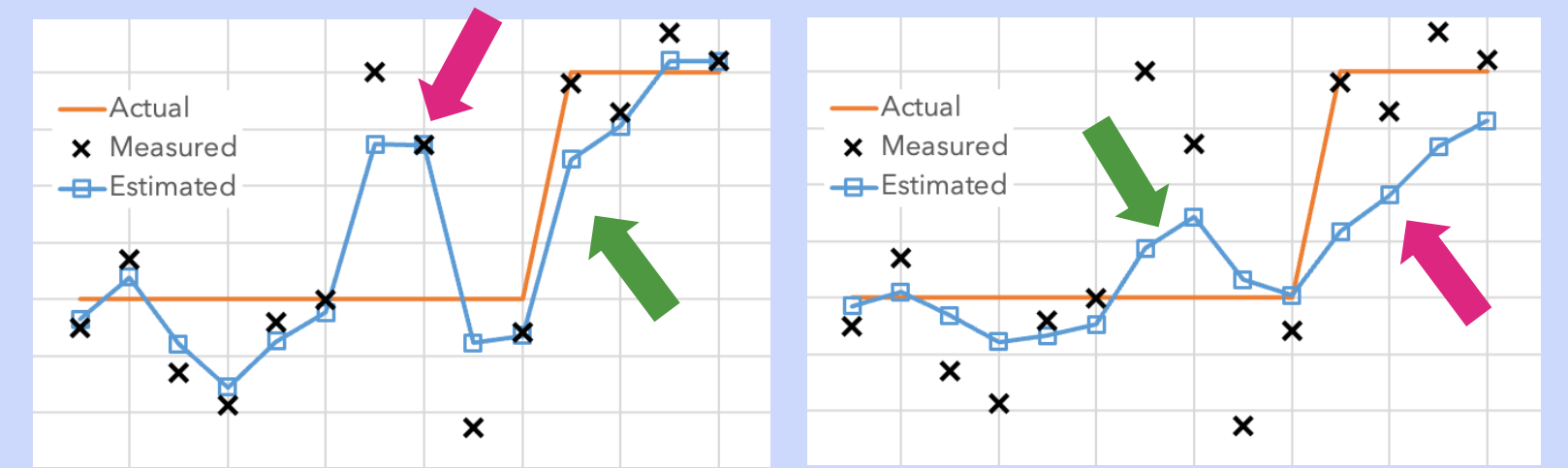
Kalman filter state update equation

$\hat{x}_{k,k}$		$\hat{x}_{k,k-1}$		K_k		z_k		$\hat{z}_{k,k-1}$
Estimate of current state	=	Predicted value of current state	+	Factor	× (	Measurement	-	Predicted value of measurement)
		from previous iteration		Kalman gain		Measurement residual Innovation (new information from present measurement)		

$$\hat{x}_{k,k} = \hat{x}_{k,k-1} + K_n \times (z_k - \hat{z}_{k,k-1})$$

State Update Equation

Kalman gain is a function of uncertainty estimates



Emphasize new measurement

Emphasize previous estimate

With gratitude to: https://www.kalmanfilter.net/

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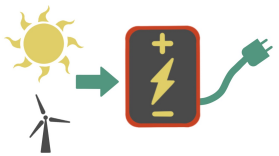
https://ofb.net/~ania/teaching/BMS-course/

Kalman filter state update equation

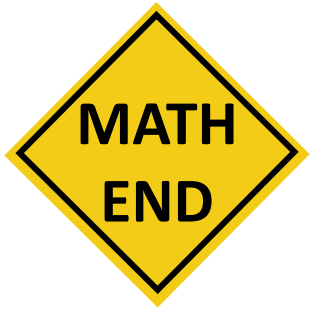


$\hat{x}_{k,k}$		$\hat{x}_{k,k-1}$		K_k		z_k		$\hat{z}_{k,k-1}$	
Estimate of current state	=	Predicted value of current state	+	Factor	× (	Measurement	-	Predicted value of measurement	)
		from previous iteration		Kalman gain		Measurement residual Innovation (new information from present measurement)			

$$\hat{x}_{k,k} = \hat{x}_{k,k-1} + K_k \times (z_k - \hat{z}_{k,k-1})$$

State Update Equation

$$K_k = \frac{uncertainty\ of\ estimate}{(uncertainty\ of\ estimate) + (uncertainty\ of\ measurement)} = \frac{p_{k,k-1}}{p_{k,k-1} + r_k}$$



With gratitude to: https://www.kalmanfilter.net/

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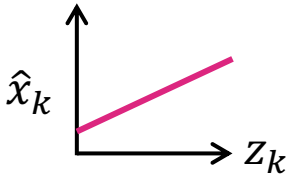
Kalman filter is part of a family

Kalman Filter (KF) and its family: Methods for estimating unobservable state variables from observable variables that may contain measurement errors.

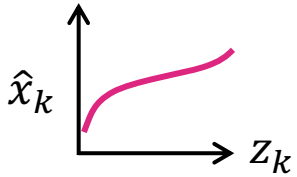
Ex. cell voltage and pack current

Ex. SOC

Kalman Filter (KF): Assumes a linear system.



Extended Kalman Filter (EKF): Extended to work for non-linear systems. Most common. O[n^3]



Sigma-point (unscented) Kalman Filter (SPKF/UKF): Conceptually more complicated, and relies on concepts often not taught to undergraduates. Same computational complexity (O[n^3]) as EKF, but SPKF has a larger constant of proportionality so requires some more floating point operations than EKF.

Particle filter: More accurate than EKF and SPKF/UKF, but generally more computationally expensive. Different internally: Kalman filters use a Taylor-series expansion, while particle filters use function evaluations and don't assume Gaussian distribution.

Central Difference Kalman Filter (CDKF): Easy to tune with only one tuning parameter *h*.

For details in math form: ECE5720 by Prof. Plett http://mocha-java.uccs.edu/ECE5720/ or https://www.coursera.org/instructor/gplett

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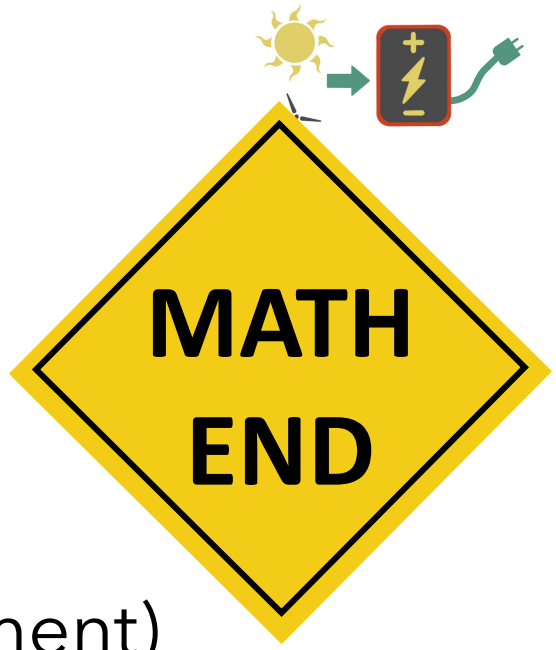
Kalman filter key concepts



The Kalman filter is:

Iterative

Embeds System Model (ex. Cell Model)

Designed to work with noise (process & measurement)

Further resources

Intuition: https://www.kalmanfilter.net/ by Alex Becker

All the math for Kalman filters applied to BMS by Prof. Plett:

- ECE5720, http://mocha-java.uccs.edu/ECE5720/
- https://www.coursera.org/instructor/gplett

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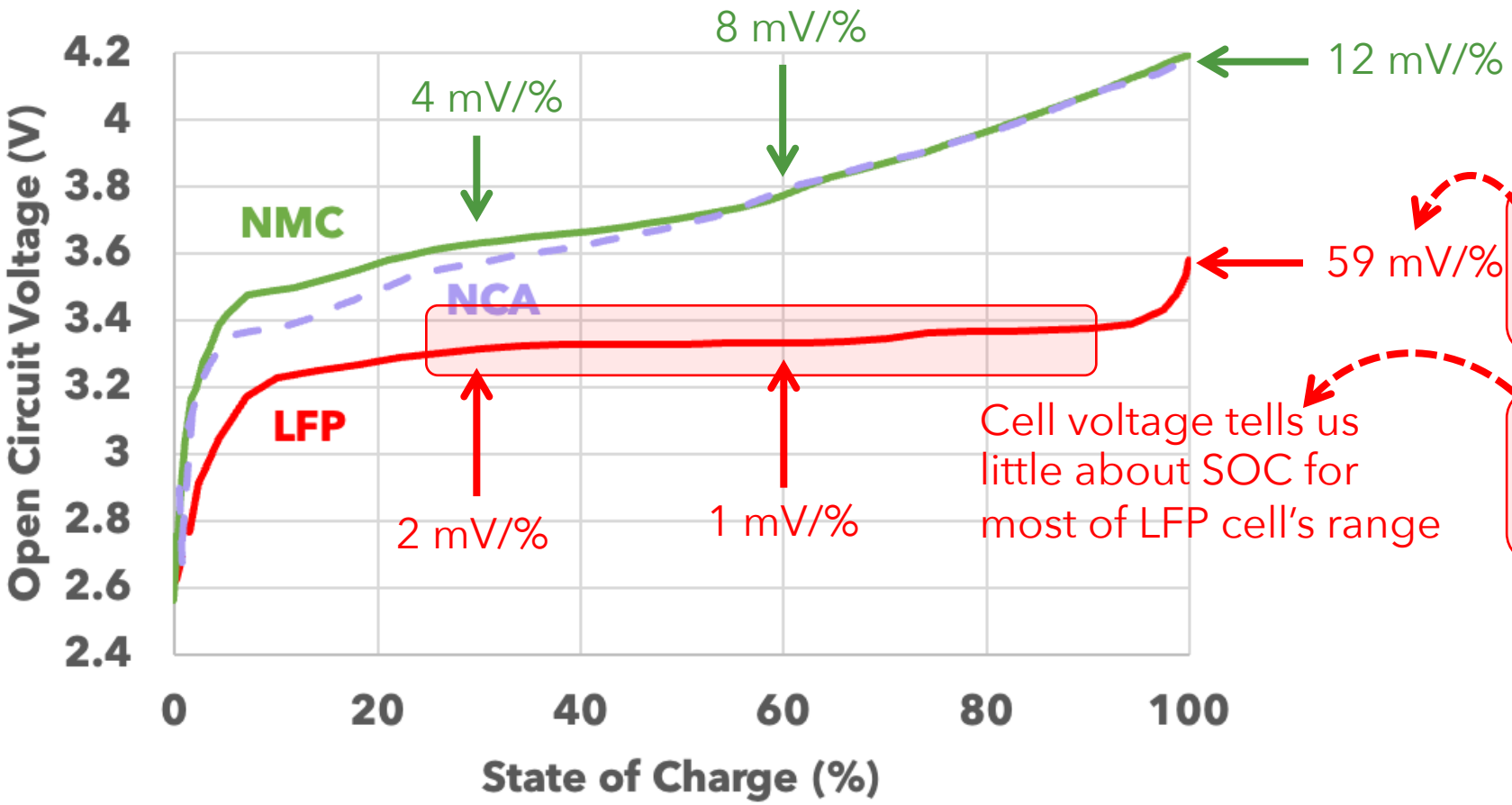
OCV-SOC slope matters for SOC estimation

Voltage is more useful with steep OCV-SOC slope



Safety
(Don't over-charge)
Easy with V_{CELL}

SOC estimation
Hard with V_{CELL}

Data from Fig. 1 in DOI: 10.1149/2.0411609jes [Keil 2016]

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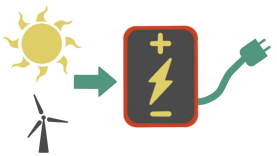
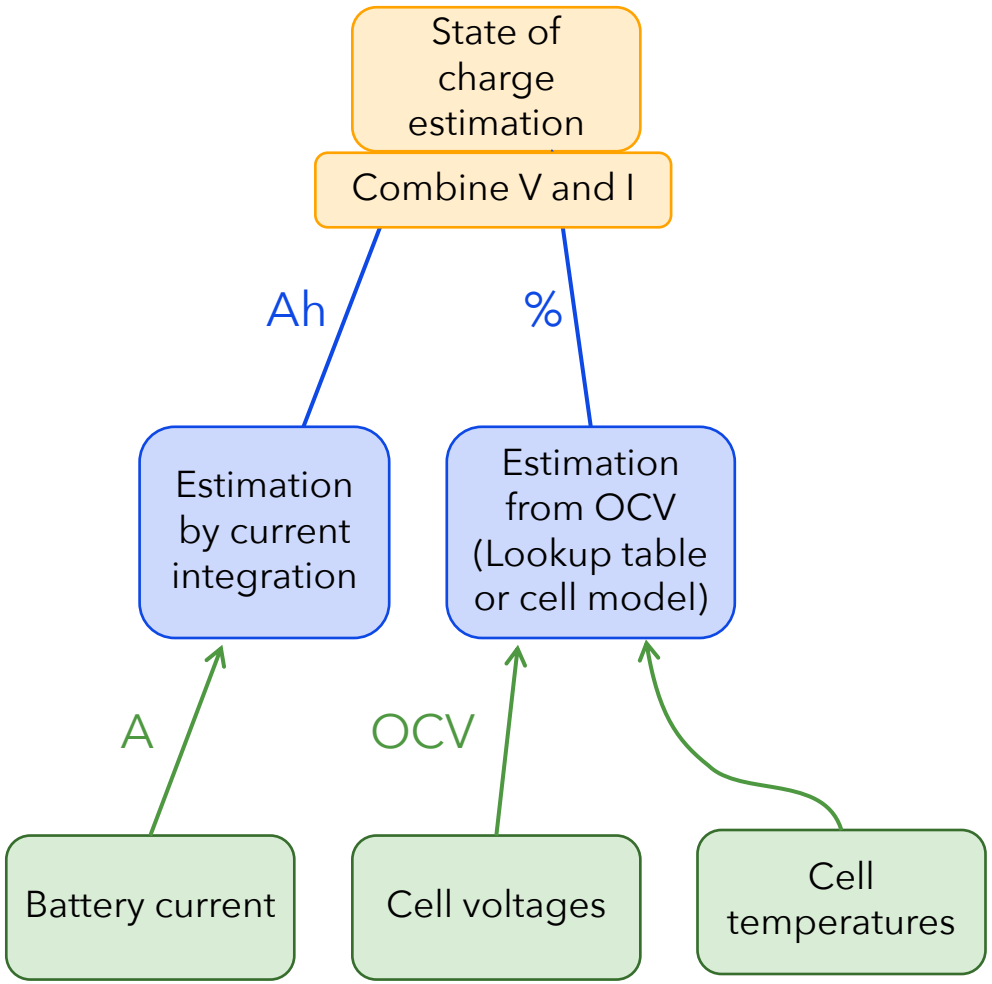
State of charge (SOC) estimation summary





- **Battery current** provides best information during charge/discharge
- **Cell voltage** provides ground truth – ultimate limit on SOC estimation accuracy
- **Combining** cell voltage and battery current information is powerful
- Methods for **sensor fusion** provide more optimal results than simpler algorithms. Extended Kalman filters are an example of an iterative model-based method for sophisticated SOC estimation.
- **Slope of OCV-SOC** curve affects usefulness of voltage information
- **Safety** and **SOC estimation** may entail different hardware requirements

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BMS State Estimation Algorithms

Just the basics.

- Overview
- State of charge
- State of health ← Capacity fade
- Power estimation

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Review: Aging and capacity fade

Capacity degrades with calendar time…

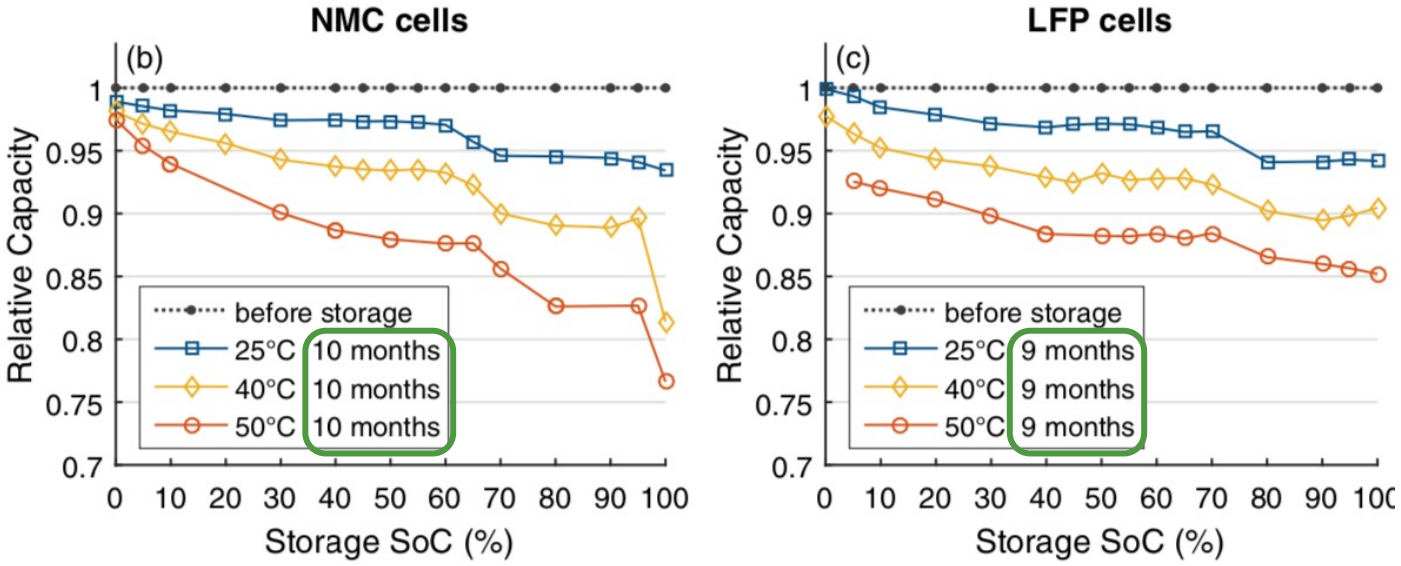
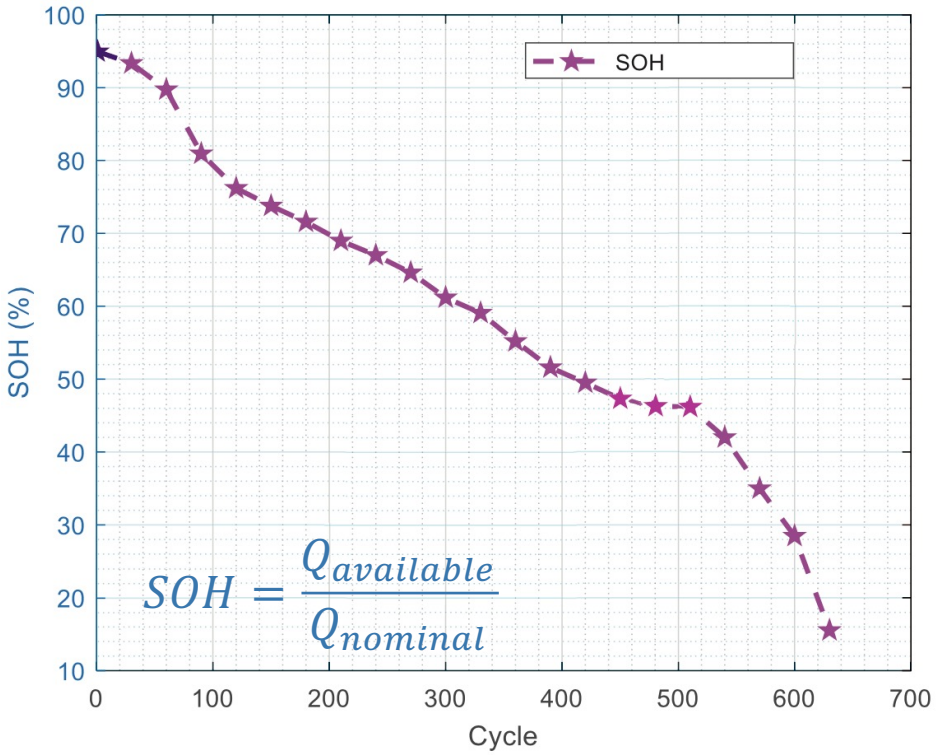


Fig. 2 in DOI: 10.1149/2.0411609jes [Keil 2016]

…and with cycling



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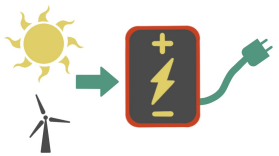
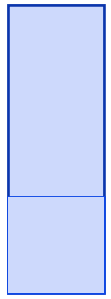
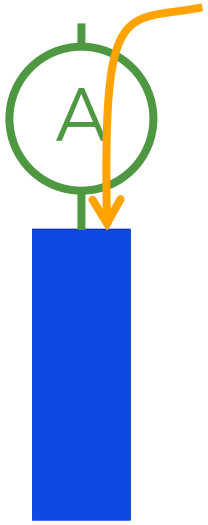
What a cell cycler does





1) Discharge to 0%
(stop at min operating voltage)



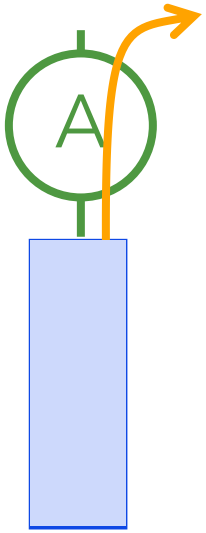
2) Charge to 100%, measuring current very accurately.
(stop at max operating voltage)

Thus, total capacity is also very accurate

Very accurate

$$Q_{BAT} = \int I_{BAT} \cdot dt$$

3) Sum the Amps



4) Discharge to 0%, measuring current very accurately

$$Q_{BAT} = \int I_{BAT} \cdot dt$$

5) Sum the Amps

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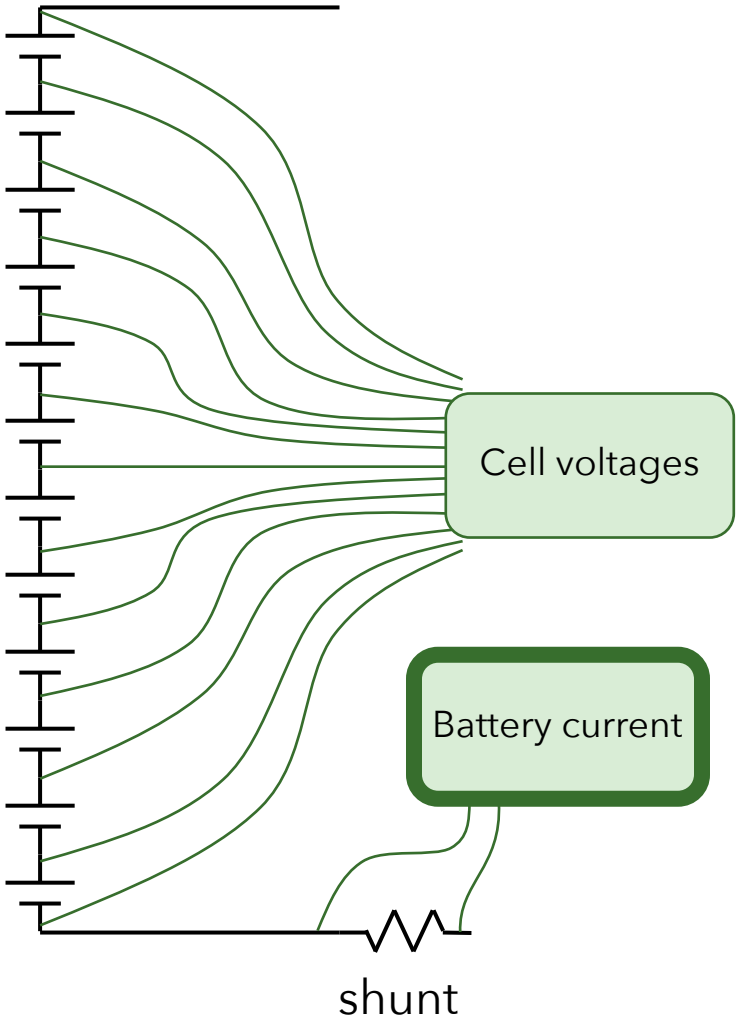
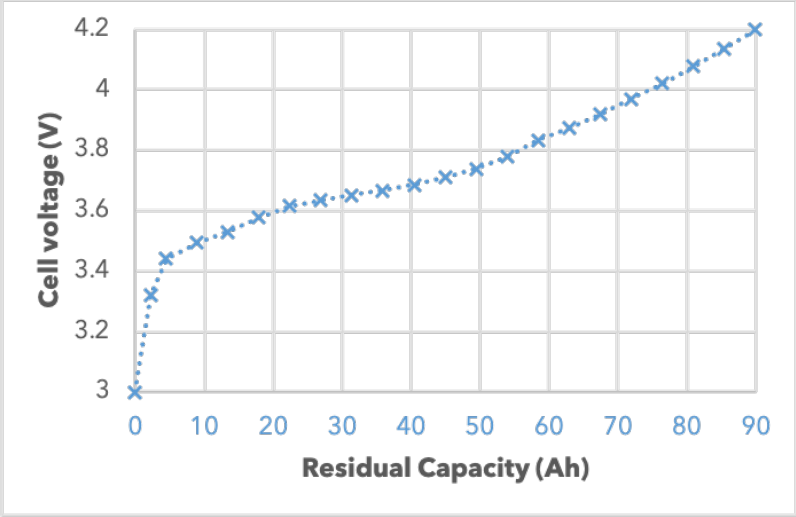
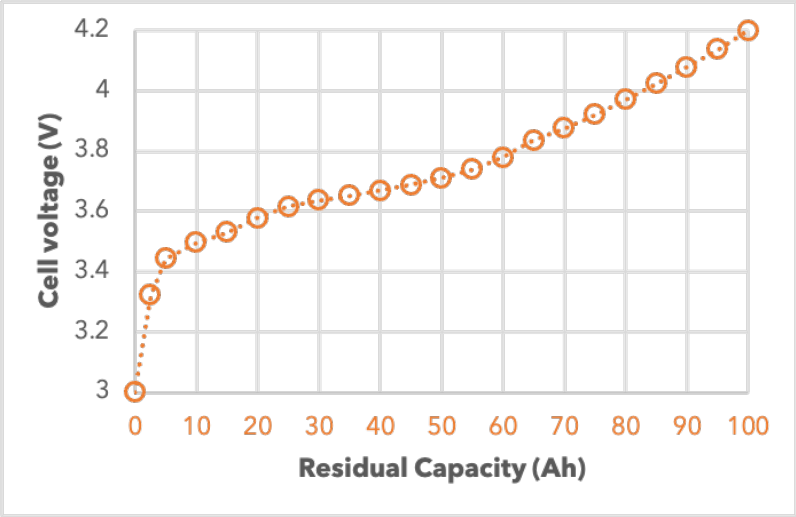
How can BMS assess present total capacity?







$$Q_{BAT} = \int I_{BAT} \cdot dt$$

Need battery current measurement

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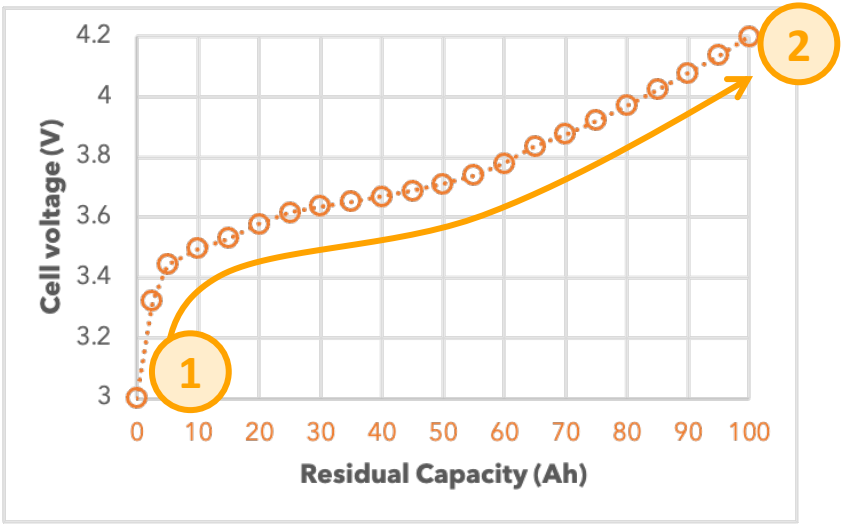
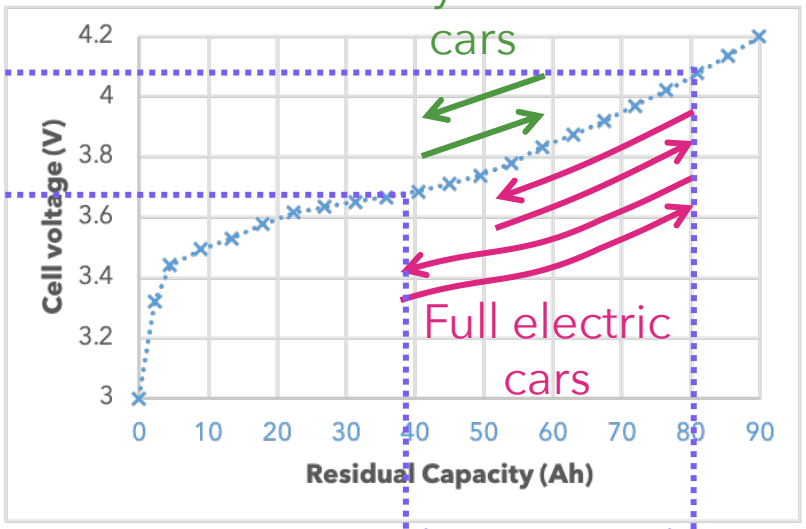
In-the-field total capacity estimation



1) Start at 0%

2) Charge to 100%, measuring current

$$Q_{BAT} = \int I_{BAT} \cdot dt$$



SOC estimate from OCV

ΔSOC (ex. 51%)

ΔQ_{BAT} (ex. 46Ah) from current sensor

$$Q_{BAT} = \frac{\Delta Q_{BAT}}{\Delta SOC} = \frac{46Ah}{51\%} = 90Ah$$

$$\Delta Q_{BAT} = \int I_{BAT} \cdot dt$$

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Error of in-the-field total capacity estimation

Error of:

$$Q_{BAT} = \frac{\Delta Q_{BAT}}{\Delta SOC}$$

1) Current sensor error
   - Directly proportional

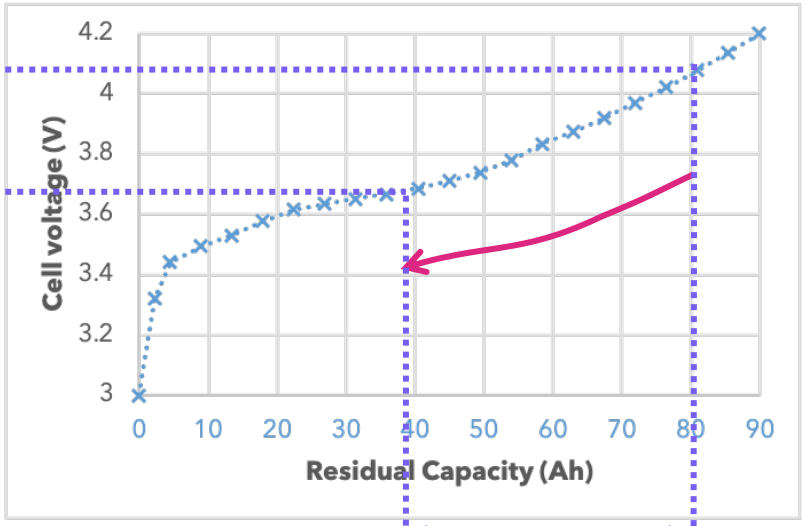
$$Q_{BAT} = \frac{\Delta Q_{BAT}}{\Delta SOC} = \frac{\int I_{BAT} \cdot dt}{\Delta SOC}$$

2) Time base error
   - Directly proportional

$$Q_{BAT} = \frac{\Delta Q_{BAT}}{\Delta SOC} = \frac{\int I_{BAT} \cdot dt}{\Delta SOC}$$

3) Cell voltage error
   - Non-linear relationship

$$Q_{BAT} = \frac{\Delta Q_{BAT}}{\Delta SOC}$$



SOC estimate from OCV

ΔSOC (ex. 51%)

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$$\Delta Q_{BAT} = \int I_{BAT} \cdot dt$$

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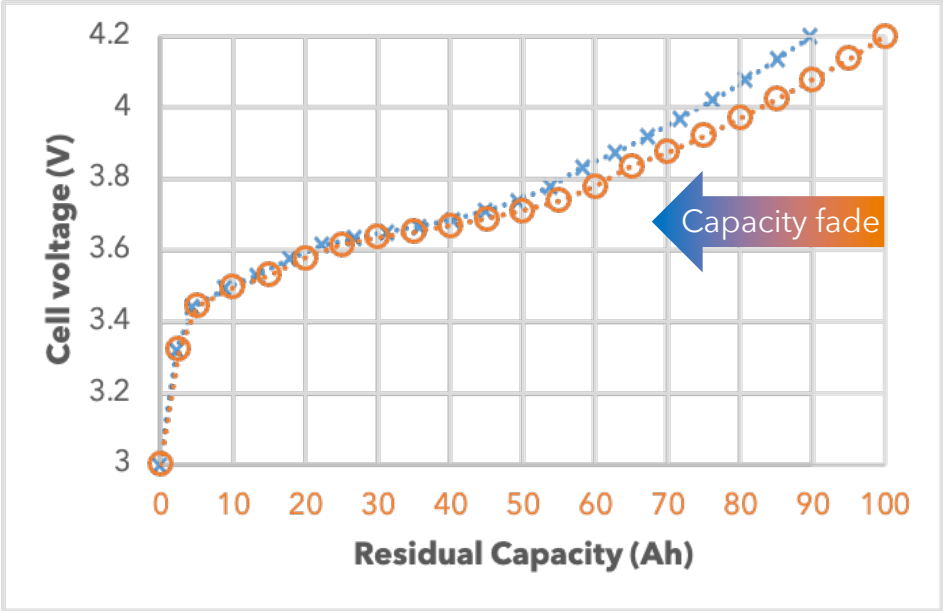
State of health (total capacity) estimation

SOH is often defined as:

$$SOH = \frac{Q_{BAT}}{Q_{nominal}}$$



Battery current measurement

Time measurement

$$\Delta Q_{BAT} = \int I_{BAT} \cdot dt$$

$$Q_{BAT} = \frac{\Delta Q_{BAT}}{\Delta SOC}$$

Cell voltage measurement → SOC estimate from OCV → ΔSOC

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BMS State Estimation Algorithms

Just the basics.

- Overview
- State of charge
- State of health
- Power estimation ← Increase in cell impedance

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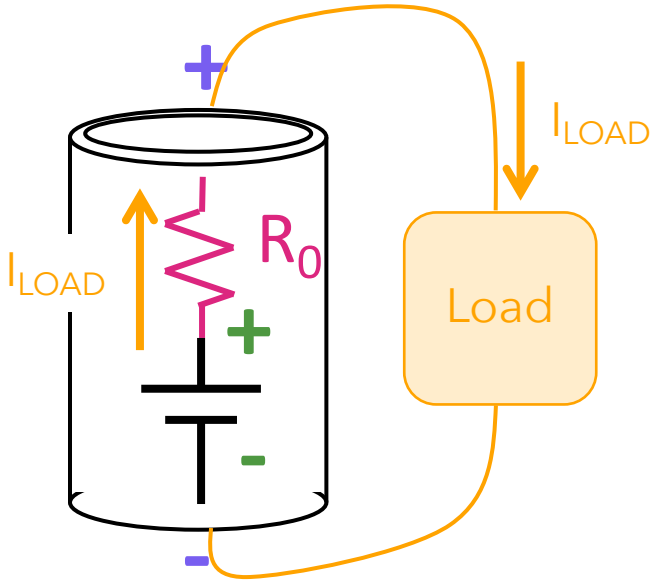
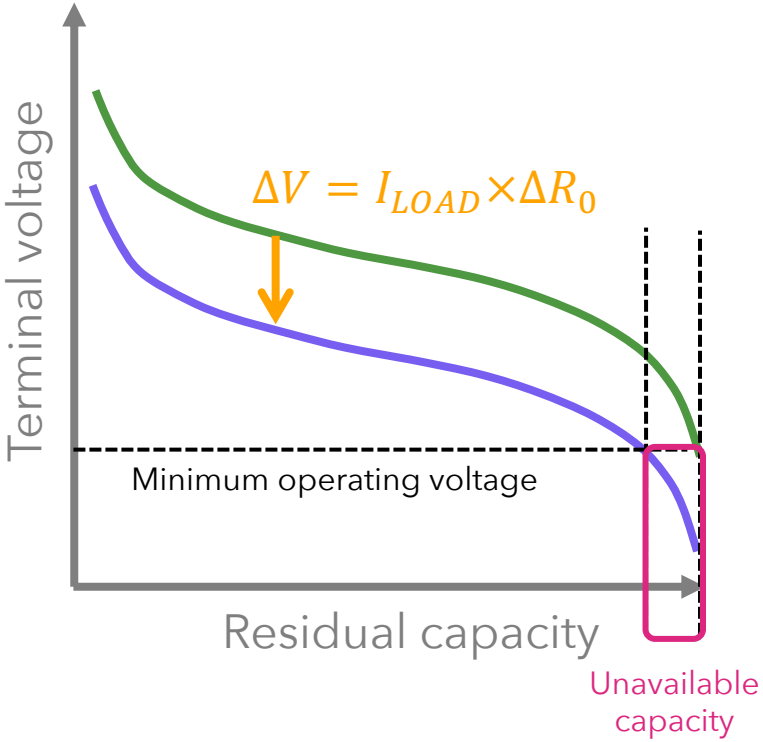
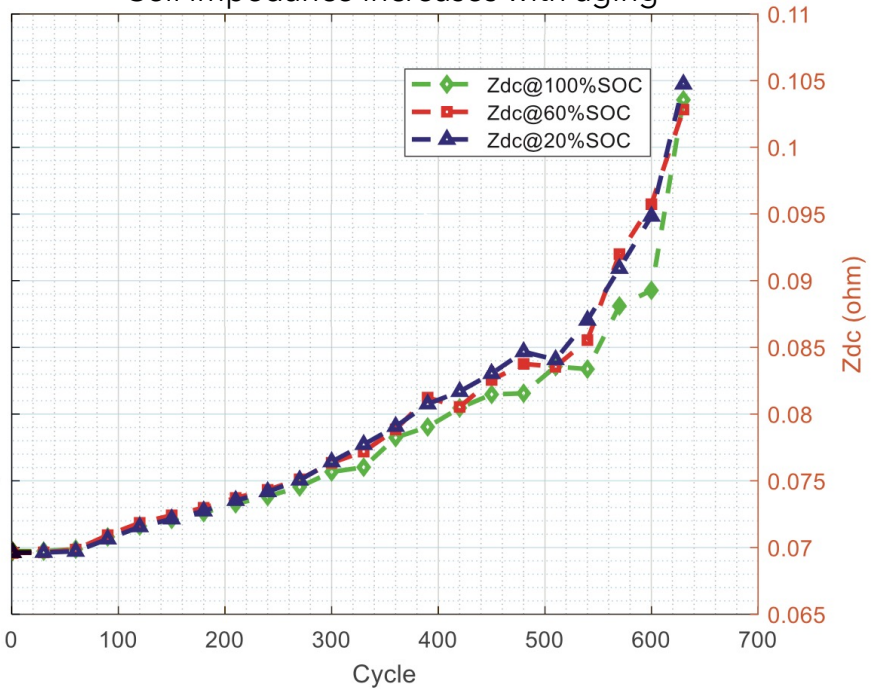
Cell impedance review

Used for available power estimation





Cell impedance increases with aging



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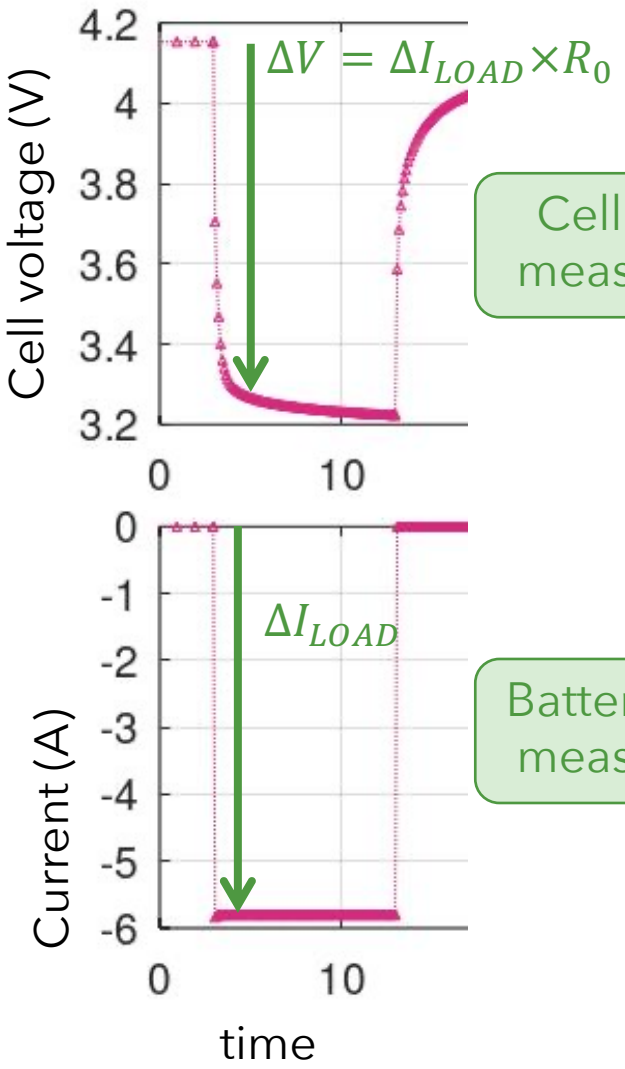
Estimating cell impedance R_0



Cell voltage measurement

Battery current measurement

$$R_0 = \frac{\Delta V}{\Delta I_{LOAD}} = \frac{(V_k - V_{k-1})}{(I_{k-1} - I_k)}$$

Measurements at time k and at time k-1

Note that R_0 varies with

- temperature
- SOC

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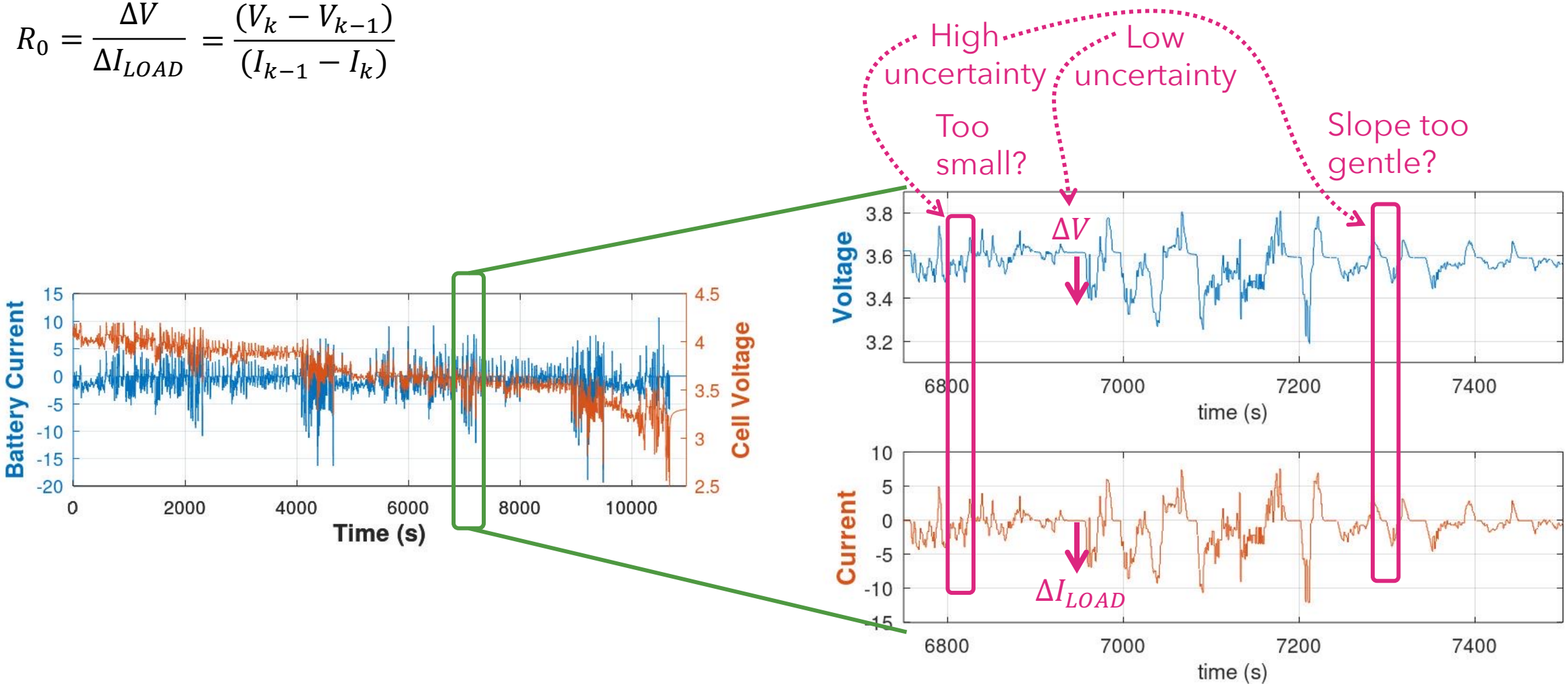
Estimating cell impedance R_0 in practice

$$R_0 = \frac{\Delta V}{\Delta I_{LOAD}} = \frac{(V_k - V_{k-1})}{(I_{k-1} - I_k)}$$



Data from Dr. Phillip Kollmeyer (phillip.kollmeyer@gmail.com)
https://data.mendeley.com/datasets/wykht8y7tq/1#file-57a2002f-d618-4dd8-96e3-37ffb6bc26c5

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"The End" by Gerd Altmann from Pixabay